Isabel Chen

(+886)0901-111-631 | tw.isabel @gmail.com | linkedin.com/in/isabelchen1 | github.com/issy16

EDUCATION

University of California, Riverside

B.S. Computer Science **June 2025, GPA 3.98/4.00 (Core 4.00)**

AI, Machine Learning, Data Mining, Big Data, Discrete Structures, Probability & Statistics, Computer Security, DBMS, GPU Architecture, Parallel Programming, Mobile Wireless Networks

Languages & Frameworks: *C++, Java, Python, SQL, JavaScript, React, HTML/CSS*

Tools & Systems: *Git, Linux/Unix, SQL Server, OutSystems, REST APIs, Scrum*

Data & ML: *Spark, Hadoop, Pandas, Scikit-learn, NoSQL, LLM APIs, ETL Pipelines, ML Pipelines*

WORK EXPERIENCE

Software Developer – Ringus Solution Enterprise Limited ***Dec 2025 – Oct 2026***

- Developed modules for a large-scale government budgeting and financial management system (OutSystems, SQL Server, JavaScript), covering budget planning, allocation, and financial reporting.
- Implemented multi-stage financial workflows with role-based access control, approval routing, and state transitions.
- Designed and optimized complex SQL queries for budget version comparison and financial reporting, handling relational data, sequential calculations, and performance bottlenecks.
- Built bulk-import and data synchronization workflows with validation, parent-child record mapping, error handling, and cross-module consistency checks.

ML Research Intern NLP Focus – UC Riverside CS Department [link] ***Sep 2024 – Dec 2024***

- Built a full ML pipeline using Google Gemini and BERTopic to classify posts by topic and severity.
- Mined 40K+ Reddit posts to uncover student stressors from academic burnout to social isolation.
- Reduced manual analysis by 90% via NLP-based topic surfacing, streamlining research review.

Data Science Intern – Monet Analytics ***Jun 2023 – Aug 2023***

- Diagnosed regression model failures through analytical problem-solving, reducing prediction error by 20%.

Undergraduate Course Grader – UC Riverside CS Department ***Apr 2024 – Jun 2025***

Discrete Structures (2025), Database Management (2024)

- Assessed 200+ upper-division CS assignments in SQL, relational algebra, and mathematical proofs involving logic, combinatorics, and graph theory.
- Collaborated with faculty using strong verbal and written communication skills to refine rubrics and identify common student misconceptions.

PROJECTS

AI-Powered WhatsApp Timeline Summarizer ***June 2025 – Jul 2025***

- Developed a full stack Chrome extension (JavaScript, CSS) inspired by a personal need for conversation insights.
- Integrated backend service logic with LLM APIs to generate AI-powered chat summaries.
- Enabled users to browse WhatsApp chats by month/year and generate AI summaries.

AI Optimization for Port Container Operations [link] ***Sep 2024 – Dec 2024***

- Developed the entire A*-based backend powering container placement.
- Designed heuristic functions to cut repositioning steps by 45% in simulation.
- Led requirements gathering and stakeholder sessions.

Machine Learning Appendicitis Detection (Python, Scikit-Learn, NumPy) ***Jan 2024 – Mar 2024***

- Built ML classifiers for pediatric appendicitis using real-world patient data.
- Led a team of 4 to develop models achieving 97% accuracy on 800+ records.

AWARDS

UCR Rose Hack 2024 Winner — Machine Learning and Engineering (Python) [link]

Built a sorting robot's ML pipeline, enabling real-time classification with 95% accuracy.

Summa Cum Laude, University of California, Riverside — Class of 2025